STA2002 Tutorial 3

1. Thirteen tons of cheese, including "22-pound" wheels (label weight), is stored in some old gypsum mines. A random sample of $n = 9$ of these wheels yielded the following weights in pounds:

21.50	18.95	18.55	19.40	19.15
22.35	22.90	22.20	23.10	

(a) Assuming that the distribution of the weights of the wheels of cheese is $N(\mu, \sigma^2)$, write down the distribution of $\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\frac{(n-1)S^2}{\sigma^2}$ where $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$.
(b) Find a two-sided 95% confidence interval for μ if we know $\sigma = 2$. Find a two-sided 95% confidence interval for μ if σ is unknown.
($z_{0.025} = 1.960, z_{0.05} = 1.645, t_{0.025}(8) = 2.306, t_{0.05}(8) = 1.860$)

Solution:

(a)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

(b) When we know $\sigma = 2$, we have $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. Then a two-sided $(1-\alpha)100\%$ C.I. for μ is given by

$$\left[\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right].$$

Substitute $\bar{x} = 20.9$ and $z_{0.025} = 1.96$, we obtain a two-sided 95% C.I. for μ is

$$\left[20.9 - 1.96 \times \frac{2}{\sqrt{9}}, 20.9 + 1.96 \times \frac{2}{\sqrt{9}}\right] = [19.59, 22.21].$$

When σ^2 is unknown, we use the sample variance S^2 to estimate it, we have $s = 1.86$, then

$$\frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2}/(n-1)}} = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

Thus, a two-sided $(1-\alpha)100\%$ C.I. for the mean when the variance is unknown is

$$\bar{x} \pm t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}$$

A 95% confidence interval for μ is

$$\left[20.9 - 2.306 \times \frac{1.86}{\sqrt{9}}, 20.9 + 2.306 \times \frac{1.86}{\sqrt{9}}\right] = [19.47, 22.33]$$

2. Let X and Y equal, respectively, the blood volumes in milliliters for a male who is a paraplegic and participates in vigorous physical activities and for a male who is able-bodied and participates in everyday, ordinary activities. Assume that $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. Following are $n = 7$ observations of X :

$$1612 \quad 1352 \quad 1456 \quad 1222 \quad 1560 \quad 1456 \quad 1924$$

Following are $m = 10$ observations of Y :

$$\begin{array}{ccccc} 1082 & 1300 & 1092 & 1040 & 910 \\ 1248 & 1092 & 1040 & 1092 & 1288 \end{array}$$

Use the observations of X and Y to
(a) Give a point estimator for $\mu_X - \mu_Y$
(b) Find the 95% confidence interval for $\mu_X - \mu_Y$ if we assume X and Y have equal variance. What if they do not have equal variance?

Solution:

(a) Give a point estimator for $\mu_X - \mu_Y$

$$\bar{x} - \bar{y} = 1511.714 - 1118.400 = 393.314$$

(b) - If the variances σ_X^2 and σ_Y^2 are equal, we use pooled T. We have

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(m + n - 2).$$

We have $S_p^2 = 29046.52$. Thus the 95% C.I. for $\mu_X - \mu_Y$ is [214.2956, 572.3324].

- If the variances σ_X^2 and σ_Y^2 are not equal, we use Welch's T. End points for confidence interval:

$$(\bar{x} - \bar{y}) \pm t_{0.025}(r)\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$$

where $s_x^2 = 49,669.905, s_y^2 = 15,297.600$. Degree of freedom:

$$r = \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{s_x^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{s_y^2}{m}\right)^2}$$

where $r = [8.599] = 8, t_{0.025}(8) = 2.306$. So the confidence interval is $[179.148, 607.480]$.

3. A candy manufacturer selects mints at random from the production line and weighs them. For one week, the day shift weighed $n_1 = 194$mints and the night shift weighed $n_2 = 162$mints. The numbers of these mints that weighed at most 21 grams was $y_1 = 28$ for the day shift and $y_2 = 11$ for the night shift. Let p_1 and p_2 denote the proportions of mints that weigh at most 21 grams for the day and night shifts, respectively.
(a) Find the endpoints for a 95% confidence interval for p_1
(b) Determine the sample size n so that we are 90% confident that the maximum error of the estimate of p_1 is $\epsilon = 0.04$. ($z_{0.05} = 1.645$)
(c) Find a one-sided 95% confidence interval that gives a lower bound for $p_1 - p_2$

Solution:

(a) Find the endpoints for a 95% confidence interval for p_1 End points:

$$\hat{p}_1 \pm z_{0.05/2}\sqrt{\frac{\hat{p}_1\left(1-\hat{p}_1\right)}{n_1}}$$

C.I.

$$0.144 \pm 1.96\sqrt{(0.144)(0.856)/196} \rightarrow [0.095, 0.193]$$

(b) Since

$$n = \frac{z^2_{\alpha/2}\hat{p}(1-\hat{p})}{\epsilon^2} \quad \text{where} \quad \epsilon = 0.04, p_1 = 0.144, z_{\alpha/2} = 1.645$$

Therefore, $n = 208.4722 \approx 209$
(c) Find a one-sided 95% confidence interval that gives a lower bound for $p_1 - p_2$ Lower bound:

$$(\hat{p}_1 - \hat{p}_2) - z_{0.05}\sqrt{\frac{\hat{p}_1\left(1-\hat{\rho}_1\right)}{n_1} + \frac{\hat{\rho}_2\left(1-\hat{p}_2\right)}{n_2}}$$

C.I.

$$\left[0.076 - 1.645\sqrt{\frac{(0.144)(0.856)}{194} + \frac{(0.068)(0.932)}{162}}, 1\right] \rightarrow [0.024, 1]$$

4. Theorem: If $X_1, X_2, \ldots, X_n$ are normally distributed and $a = \chi^2_{1-\alpha/2,n-1}$ and $b = \chi^2_{\alpha/2,n-1}$, then a $(1-\alpha)\%$ confidence interval for the population variance σ^2 is:

$$\left(\frac{(n-1)s^2}{b} \leq \sigma^2 \leq \frac{(n-1)s^2}{a}\right)$$

And a $(1-\alpha)\%$ confidence interval for the population standard deviation σ is:

$$\left(\frac{\sqrt{(n-1)}}{\sqrt{b}}s \leq \sigma \leq \frac{\sqrt{(n-1)}}{\sqrt{a}}s\right)$$

We learned previously that if $X_1, X_2, \ldots, X_n$ are normally distributed with mean μ and population variance σ^2, then:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

with $\left(a = \chi^2_{1-\alpha/2}\right)$ and $\left(b = \chi^2_{\alpha/2}\right)$, we can write the following probability statement:

$$P\left[a \leq \frac{(n-1)S^2}{\sigma^2} \leq b\right] = 1-\alpha$$

Now, as always it's just a matter of manipulating the quantity in the parentheses. That is:

$$a \leq \frac{(n-1)S^2}{\sigma^2} \leq b$$

Taking the reciprocal of all three terms, and thereby changing the direction of the inequalities, we get:

$$\frac{1}{a} \geq \frac{\sigma^2}{(n-1)S^2} \geq \frac{1}{b}$$

Now, multiplying through by $(n-1)S^2$, and rearranging the direction of the inequalities, we get the confidence interval for σ^2 :

$$\frac{(n-1)S^2}{b} \leq \sigma^2 \leq \frac{(n-1)S^2}{a}$$

as was to be proved. And, taking the square root, we get the confidence interval for σ :

$$\frac{\sqrt{(n-1)S^2}}{\sqrt{b}} \leq \sigma \leq \frac{\sqrt{(n-1)S^2}}{\sqrt{a}}$$

as was to be proved.